

# Wireless Communication Exam Notes

EX 715 (BEI) / EX 751 (BEX) · IV-I · TU, IOE · tiers over 23 papers, 2070–2082 BS

## 2. Cellular Mobile Communication Concept

4 hrs · asked in **21 of 23 papers** · 5 syllabus sub-topics · **228 marks, 11.2% of the archive** · heavy numerical load → see the Ch 2 numerical companion.

### Read this first:

- Two questions carry the chapter. **Handoff** **TOP 8/23** and the **proof**  $Q = \sqrt{3N}$  **TOP 6/23**. Bank both and most papers are already covered.
- Everything in this chapter hangs off **one trade-off**: small  $N \rightarrow$  more capacity but worse SIR. Large  $N \rightarrow$  better SIR but less capacity. Say that sentence in any answer here and it will fit.
- The examiner reliably pairs a theory part with a numerical part (**prove**  $Q = \sqrt{3N}$ , **then find optimal  $N$  for a given SIR**). Do not learn the proof without the numerical.

### 2.1 The cellular concept

**What is the cellular concept? What is a cell footprint? Explain interference tier** **1+4 5**

80 Ch · background for every other question in the chapter

#### The problem it solved

- Early mobile radio = **one high-power transmitter** on a tall tower.
  - Large coverage, but the same frequencies could **never** be reused inside it.
  - Bell's New York system supported **12 simultaneous calls over 1000 square miles**.
- Spectrum is fixed by the regulator → capacity could not grow.

#### Advantages of going cellular

- Low transmit power → smaller batteries, cheaper hardware, less interference.
- Frequency reuse → capacity grows w/o new spectrum.
- Capacity can be grown later by splitting cells, w/o redesigning the system.

#### The cellular idea (3 system-level ideas)

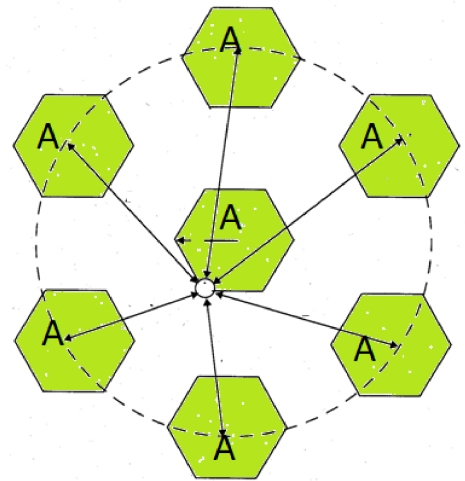
- Replace one high-power transmitter with **many low-power transmitters**, each covering a small area called a **cell**.
- Give **neighbouring cells different channel groups** → interference stays low.
- **Reuse the same channel set** at cells far enough apart.
- Result: high capacity in a fixed spectrum, **with no new technology**.

#### Why a hexagon +Fig

- The ideal contour is a **circle**, but circles either **overlap** (double coverage) or leave **gaps** (no coverage).
- Only 3 regular shapes tile a plane w/o gaps or overlap: **triangle, square, hexagon**.
- For the same distance from centre to the farthest edge, the **hexagon covers the largest area** → **fewest cells, fewest base stations, lowest cost**.
- Hexagon is also the closest of the three to a circle → best match to a real omni-directional antenna pattern.
- It has exactly **6 equidistant neighbours**, and centre-to-centre lines are separated by multiples of  $60^\circ$  → makes the reuse geometry solvable.

## Terms

- **Cell** = area covered by one cellular transmitter. The transmitter site itself = **cell site**.
- **Cell footprint** = the **actual** radio coverage of a cell.
  - Real footprints are irregular (terrain, buildings).
  - Irregular footprints are tolerable at first, but as cells are added they **break the reuse plan** and cause co-channel interference.
  - So a **regular shape is assumed** for planning.
- **Cluster** = the  $N$  cells that together use the complete channel set.
- **Co-channel cells** = cells using the same channel set as the target cell.
- **Interference tier** = a set of co-channel cells all at the **same distance** from the reference cell.
  - The nearest such set = **first tier**.
  - **In hexagonal geometry the first tier always has exactly 6 cells.**
  - Second and higher tiers are much further away, so their interference is normally neglected.



First tier: 6 co-channel cells, all at distance  $D$  from the reference cell.

## 2.2 Frequency reuse and cluster size

### Explain the frequency reuse concept and the co-channel reuse ratio 6 4 3 2+4

**HOT 4/23** 72 Ma, 74 Bh, 76 Bh, 82 Ba **2+4** · 74 Bh words it as “minimizing reuse distance maximizes spectral efficiency” · 82 Ba pairs it with the channel assignment strategies below

#### Frequency reuse / frequency planning

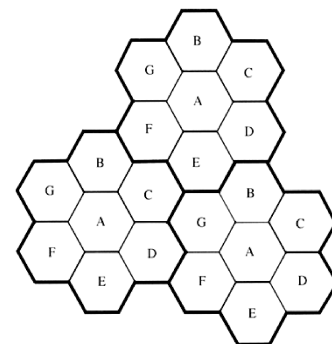
- Each BS gets a group of channels, used only inside its own cell.
- Neighbouring cells get **different** groups.
- Because coverage is confined to the cell, the **same group can be reused** at a cell far enough away.
- Design task = choose the reuse pattern so interference stays tolerable.

#### Capacity arithmetic (learn this, it is the base of every numerical)

- $S$  = total duplex channels in the system.
- $k$  = channels per cell.
- $N$  = cells per cluster (cluster size).
- $M$  = number of times the cluster is replicated.
- So  $S = kN$ , and system capacity

$$C = MkN = MS$$

- **Capacity is directly proportional to  $M$** , the number of replications.
- **Frequency reuse factor** =  $1/N$ : each cell gets  $1/N$  of the total channels.
- Typical  $N = 4, 7, 12$ .
- **Small  $N \rightarrow$  cluster repeats more often  $\rightarrow$  higher capacity**, but co-channel cells sit closer  $\rightarrow$  worse interference.



**Figure 3.1** Illustration of the cellular frequency reuse concept. Cells with the same letter use the same set of frequencies.

Cells with the same letter use the same channel set.

### Which $N$ are allowed

- Hexagons only tile for

$$N = i^2 + ij + j^2 \quad i, j = 0, 1, 2, \dots$$

- Allowed  $N$ : 1, 3, 4, 7, 9, 12, 13, 16, 19, 21, ...
- Eg  $i = 1, j = 1 \rightarrow N = 3$ .  $i = 2, j = 1 \rightarrow N = 7$ .  
 $i = 3, j = 2 \rightarrow N = 19$ .

### Channel assignment strategies 4

#### Fixed channel assignment (FCA)

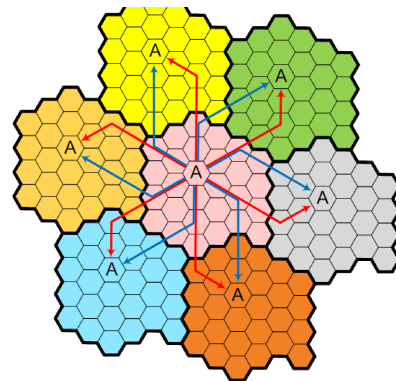
- Each cell gets a **predetermined** set of voice channels.
- A new call can only use an unused channel **of that cell**.
- If all of the cell's channels are busy, the call is **blocked**, even when a neighbour is idle.
- Simple, no signalling overhead, predictable.
- Borrowing** variation: a busy cell may borrow a channel from a neighbour, supervised by the MSC, provided it does not break the reuse plan.

### Locating the co-channel cells (use the figure above)

- From the centre of any cell:
  - move  $i$  cells perpendicular to one of the 6 faces,
  - turn  $60^\circ$ , move  $j$  cells.
- That lands on a co-channel cell. Repeating for all 6 faces gives the whole first tier.
- Clockwise and counter-clockwise turns give the two mirror routes.

#### Dynamic channel assignment (DCA)

- Channels are **not** owned by any cell. They sit in a central pool at the MSC.
- On each request the MSC allocates a channel after checking
  - the candidate channel is unused in that cell,
  - and no co-channel cell within the reuse distance is using it.
- Lowers blocking probability, raises capacity** because it follows the traffic.
- Cost: the MSC must collect occupancy and signal-strength data continuously  $\rightarrow$  heavy storage and computation load.



$i = 4, j = 2$ : the 6 nearest co-channel A cells.

## 2.3 Handoff

What is handoff? Explain the handover process and the types of handover with application 6 4+4 4

2+4 1+3

**TOP 8/23** 72 Ash, 73 Bh, 73 Ma, 74 Ma, 76 Bh, 78 Ch, 79 Bh, 81 Ba · **most repeated question in the chapter**

### Definition

- A mobile moves out of one cell into another **while a call is in progress**.
- The MSC transfers the call to a new channel belonging to the new BS.
- That transfer = **handoff** (US) = **handover** (Europe).
- Must be **transparent**: the user should not notice it.

### The handoff process, step by step

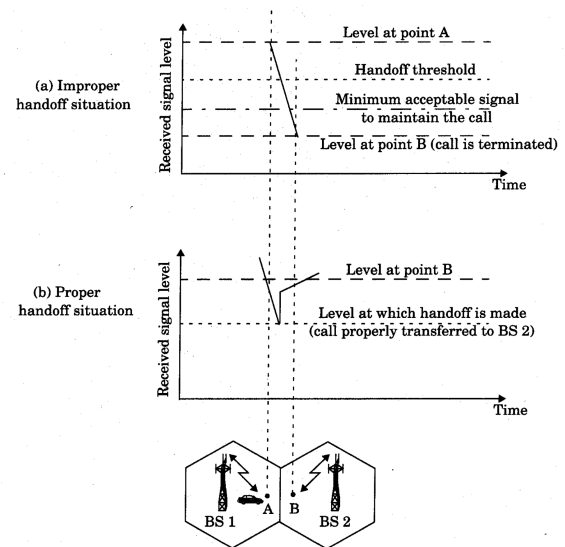
- Serving BS continuously measures the received signal strength on each **reverse voice channel** → tells the MSC roughly where each mobile is.
- Signal drops below the **handoff threshold**  $P_{r,handoff}$ .
- MSC **identifies a candidate BS** whose signal is stronger.
- MSC checks the candidate has a **free channel**.
- MSC **re-allocates voice and control channels** to the new BS.
- Call continues on the new channel, old channel returns to the pool.

### Handoff threshold and handoff margin (75 Bh asks for the margin +Fig. 3)

- $P_{r,minimum\ usable}$  = weakest signal that still gives acceptable voice, typically  $-90$  to  $-100$  dBm.
- $P_{r,handoff}$  = level at which handoff is started. Always set a little higher.
- **Handoff margin**

$$\Delta = P_{r,handoff} - P_{r,minimum\ usable}$$

- Δ too large → handoffs fire needlessly → MSC is burdened.
- Δ too small → not enough time to finish the handoff → call drops.
- 1G analog:  $\Delta \approx 6$  to  $12$  dB, handoff takes about  $10$  s.
- 2G (GSM):  $\Delta \approx 0$  to  $6$  dB, handoff takes  $1$  to  $2$  s.



Improper Vs proper handoff (76 Bh 4, 81 Ba 1+3). Top: the level is left to fall past the minimum usable value before handoff is attempted → **call terminated at B**. Bottom: handoff is made while the signal is still usable → call transferred to BS 2.

### Types of handoff

- **Hard handoff** = “break before make”.
  - Old connection is released **before** the new one is set up.
  - Used in FDMA and TDMA systems (AMPS, GSM), where the two cells are on different frequencies.
  - Risk: if the new channel fails, **the call drops**.
- **Soft handoff** = “make before break”.
  - New connection is set up **before** the old one is released, so the mobile is briefly served by two BSs at once.
  - Only possible in **CDMA**, where every cell uses the same frequency.
  - Gives **BS diversity** at the cell boundary → far fewer dropped calls.
- **Intra-cell handoff**: same BS, different channel. Used to escape interference or a deep fade.
- **Inter-cell handoff**: between two cells of the same MSC.
- **Intersystem handoff**: mobile leaves the coverage of one MSC and enters a system run by a **different MSC**. Needs an inter-MSC link and a roaming agreement.

## How the handoff decision is taken (handoff measurement)

- **1G analog:** the **BS** measures signal strength, the **MSC** decides. A spare receiver, the **locator receiver**, scans neighbouring cells' mobiles.
- **2G TDMA: mobile assisted handoff (MAHO).**
  - The **mobile** measures received power from surrounding BSs and reports continuously to its serving BS.
  - Handoff triggers when a neighbour becomes appreciably stronger.
  - **Much faster (1 to 2 s)** because the MSC no longer has to poll every BS.
  - GSM also weighs **co-channel and adjacent channel interference**, not just level.
- **IS-95 CDMA:** all cells share the channel, so **there is no physical channel change at all**. The MSC simply picks the BS receiving the strongest signal.
- In every case the decision must use a **running average** of signal strength, so that **a momentary fade is not mistaken for the mobile leaving the cell**.
  - Averaging window depends on vehicle speed.
  - A steep short-term average → handoff must be made quickly.
  - Speed can be estimated from the statistics of the short-term fading at the BS.

## Explain two schemes for prioritizing handoff 3 2

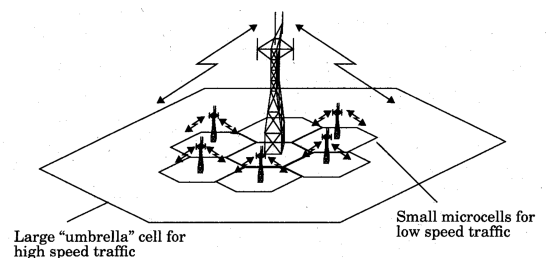
PIN 1/23 82 Bh · also asked inside 72 Ash 2

- **Why prioritize:** a **dropped call** is judged far more serious by the subscriber than a **blocked call**. So handoff requests must beat new-call requests.
- **Scheme 1: guard channels.**
  - A fixed fraction of each cell's channels is **reserved exclusively** for handoff requests.
  - Simple, immediate.
  - Cost: those channels sit idle for new calls → **total carried traffic falls**.
  - Fixed by using **dynamic channel assignment** for the reserved pool.
- **Scheme 2: queuing of handoff requests.**
  - A handoff request that finds no free channel is **put in a queue** instead of being refused.
  - Works because there is a **finite window** between the signal dropping below the handoff threshold and dropping below the minimum usable level.
  - That window leaves room to wait for a channel to free up.
  - Reduces the **probability of forced termination**, but does not eliminate it.

## Practical handoff considerations 4 4+4

HOT 4/23 79 Bh, 73 Ma, 78 Ch, 72 Ash

- **Users move at wildly different speeds.**
  - High-speed vehicle users need **frequent** handoffs.
  - Pedestrians may need **none at all** in a whole call.
- **Microcells make it worse.** Small cells raise capacity, but the MSC is swamped when fast users are handed between them every few seconds.
- **Umbrella cell** is the standard fix +Fig.
  - A large, high, high-power cell is **overlaid** on a group of microcells at the same site.
  - **High-speed** traffic is put on the umbrella cell → few handoffs.
  - **Low-speed** traffic is put on the microcells → high capacity.
  - Speed is estimated from the rate of change of the short-term signal average.



Umbrella cell: one large cell for fast traffic sitting over the microcells that serve slow traffic.

- **Cell dragging** (72 Ash, 73 Bh ask for this by name, 2 ).
  - A **pedestrian** with a clear LOS to the BS keeps a **very strong** signal.
  - The level never falls to the handoff threshold, so no handoff is triggered.
  - The user **travels deep into a neighbouring cell** while still held by the old BS.
  - Result: **severe interference in the neighbouring cell, and the reuse plan is broken.**
  - Fixed by tuning the handoff threshold and averaging window carefully.
- **Handoff requests outrank new calls** so that GoS is preserved.

### Handoff strategy used in GSM (70 Ma, 8 )

- GSM uses **MAHO**, so the mobile does the measuring.
- In every idle timeslot the mobile measures the **BCCH** power of up to **16 neighbouring cells** and reports the best 6 to its BS twice per second.
- Decision uses **RXLEV** (level) and **RXQUAL** (bit error rate), not level alone.
- So GSM will hand a call off for **poor quality even when the level is high**, which a level-only 1G system cannot do.
- Handoff is **hard** (break before make) and completes in 1 to 2 s.
- Four cases: intra-cell, inter-cell within one BSC, between two BSCs of one MSC, and between two MSCs.

## 2.4 Interference and system capacity

### Explain co-channel interference and adjacent channel interference 2.5+2.5 3 4

80 Ch, 70 Bh, 81 Bh, 80 Ba, 81 Ba · 70 Bh and 81 Bh pair it with the  $Q = \sqrt{3N}$  proof

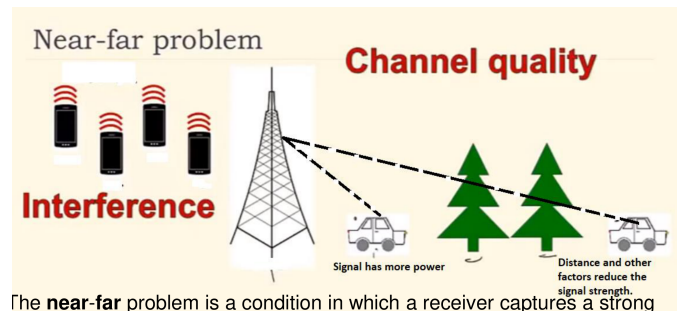
#### Sources of interference

- Another mobile **in the same cell**.
- A call in progress in a **neighbouring cell**.
- Other **base stations** operating in the same frequency band.
- Any **non-cellular system** leaking energy into the cellular band.
- Interference is **the limiting factor on capacity**, not noise.
- On **voice** channels it causes cross-talk and audible noise.
- On **control** channels it causes missed and blocked calls through signalling errors.

|                | Co-channel interference (CCI)                                     | Adjacent channel interference (ACI)                                                        |
|----------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Cause          | Another cell using the <b>same</b> frequency                      | Signal in a <b>nearby</b> frequency leaking through an imperfect receiver filter           |
| Comes from     | Co-channel cells, one reuse distance away                         | Channels next to the wanted one, often <b>inside the same cell</b>                         |
| Depends on     | $Q = D/R$ and cluster size $N$                                    | Filter sharpness and the near-far geometry                                                 |
| Independent of | <b>Transmit power</b> (raising all powers raises S and I equally) |                                                                                            |
| Cured by       | Increasing $D/R$ , ie a larger $N$ . Sectoring                    | Channel allocation w/ maximum frequency separation. Sharp band-pass filters. Power control |

## Adjacent channel interference in detail

- Receiver filters are not ideal → energy from neighbouring frequencies **leaks into the passband**.
- Near-far problem** is the worst case + Fig.
  - A mobile **close** to the BS transmits on a channel adjacent to one used by a mobile **far** from the BS.
  - The near mobile's signal arrives far stronger.
  - The BS then **cannot hear the far mobile at all**.
- Cures:
  - Channel allocation**: never give a cell channels that are adjacent in frequency. Keep separation as large as possible.
  - Careful filtering**: sharp band-pass filter at the receiver, plus modulation w/ low out-of-band radiation.
  - Separate multiplexing** for uplink and downlink.
  - Power control** (below).



The **near-far** problem is a condition in which a receiver captures a strong

### Power control syllabus, no PYQ yet

- Every mobile transmits **the smallest power** that still holds a good reverse link.
- Benefits:
  - Solves the near-far problem**: the near mobile is turned down until the far mobile is audible again.
  - Longer battery life**.
  - Raises reverse-link SIR** across the whole system.
- Essential, not optional, in CDMA**, where every user in every cell shares one radio channel, so one loud mobile can jam a whole cell.

Prove that the co-channel reuse ratio  $Q = \sqrt{3N}$ , and derive S/I 4+4 3+5 4 3+4+3 3+3

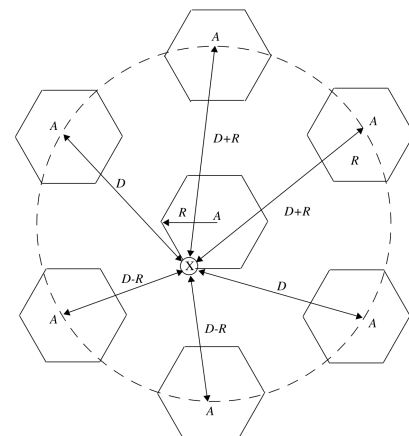
**TOP 6/23** 70 Bh, 71 Bh, 74 Ma, 79 Bh, 80 Bh, 81 Bh · **almost always followed by "find optimal N for SIR = 15 or 18 dB"** → Ch 2 numerical companion

### Step 1: what $Q$ is

- $R$  = cell radius (centre to vertex).
- $D$  = distance to the centre of the **nearest** co-channel cell.
- Co-channel reuse ratio**  $Q = D/R$ .
- When all cells are about the same size, CCI is **independent of transmit power** and depends only on  $R$  and  $D$ .
- Larger  $Q$  → less interference but less capacity**. The whole design is this trade-off.

Interference and System Capacity

71



First tier for  $N = 7$ . Worst case: the mobile sits at the cell boundary (point A), so its wanted signal is at its **minimum** while the interferers are at their nearest. Exact distances are  $D - R$ ,  $D$ ,  $D$ ,  $D$ ,  $D + R$ , but the approximation  $D_i = D$  is used for easy analysis.

**Step 2: hexagonal geometry**

- Put the cell centres on the axes  $u, v$  separated by  $60^\circ$ .
- Distance between two **adjacent** cell centres =  $\sqrt{3} R$ .
- Move  $i$  cells along  $u$ , then  $j$  cells along  $v$  at  $60^\circ$ .
- By the cosine rule w/ the included angle =  $60^\circ$ ,

$$D^2 = (i\sqrt{3}R)^2 + (j\sqrt{3}R)^2 - 2(i\sqrt{3}R)(j\sqrt{3}R)\cos 120^\circ$$

$$D^2 = 3R^2(i^2 + j^2 + ij)$$

- But  $N = i^2 + ij + j^2$ , so  $D^2 = 3NR^2$ , giving

$$D = \sqrt{3N} R \quad \Rightarrow \quad Q = \frac{D}{R} = \sqrt{3N}$$

**Step 3: signal to interference ratio**

- Received power falls as  $P_r \propto d^{-n}$ , path loss exponent  $n = 2$  to  $4$ .
- With  $N_I$  interfering cells,

$$\frac{S}{I} = \frac{S}{\sum_{i=1}^{N_I} I_i} = \frac{R^{-n}}{\sum_{i=1}^{N_I} D_i^{-n}}$$

- Take the **worst case**: mobile at the cell edge, first tier only,  $N_I = 6$ , all  $D_i = D$ :

$$\frac{S}{I} = \frac{R^{-n}}{6D^{-n}} = \frac{(D/R)^n}{6} = \frac{Q^n}{6} = \frac{(\sqrt{3N})^n}{6}$$

- Inverting for design,

$$Q = \sqrt{3N} = \left(6 \cdot \frac{S}{I}\right)^{1/n} \Rightarrow N = \frac{Q^2}{3}$$

- Then **round  $N$  up** to the next allowed  $i^2 + ij + j^2$ .

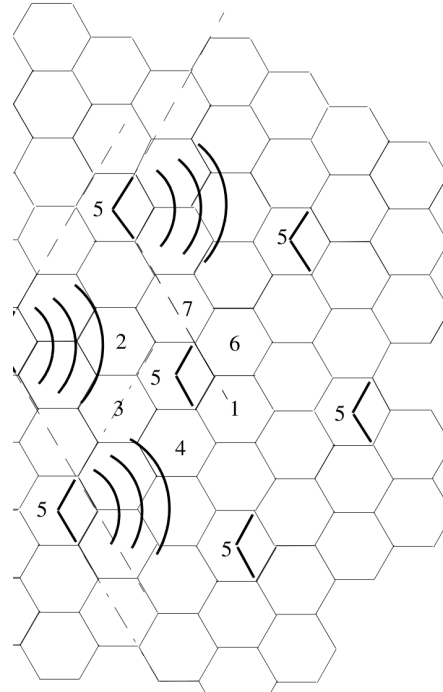
**Exact worst case for  $N = 7$ , and what sectoring does to it**

- Keeping the true distances instead of  $D_i = D$ , and  $n = 4$ :

$$\frac{S}{I} = \frac{R^{-4}}{2(D-R)^{-4} + 2(D+R)^{-4} + 2D^{-4}}$$

- For  $N = 7$ ,  $Q = 4.6$ :
  - approximate result  $\rightarrow 49.56 = 17$  dB,
  - exact result  $\rightarrow 17.8$  dB.
- Both fall **just short of the 18 dB** that AMPS demands.
- Going to the next size,  $N = 12$  ( $i = j = 2$ ), fixes SIR but cuts spectrum utilisation from  $1/7$  to  $1/12$   $\rightarrow$  **not tolerable**.
- So the answer is **sectoring**, not a bigger  $N$ .

- With  $120^\circ$  sectoring and 7-cell reuse, a mobile is hit by only **2 of the 6** first-tier co-channel cells.
- SIR rises from 17 dB to about **24.2 dB**.
- That margin is then **spent** on reducing  $N$ , which raises capacity.



120° sectoring: of the 6 first-tier co-channel cells, only 2 have antenna patterns radiating into the centre cell.

## 2.5 Trunking and grade of service

Define grade of service. How is it measured in a blocked-calls-cleared trunking system? 3 4

PIN 1/23 79 Bh · 82 Bh asks for GoS, traffic intensity, average holding time and blocked call as definitions 4 · the numerals built on this are in the Ch 2 companion

### What trunking is

- A large user population shares a **small pool of channels**.
- A channel is allocated **per call** and returned to the pool the moment the call ends.
- Works because users do not all call at once: it exploits their **statistical** behaviour.
- If every channel is busy when a user calls, that user is **blocked**.
- Trade-off: fewer channels → cheaper, but more blocking.

### Definitions (82 Bh asks for exactly these)

- **Set-up time**: time to allocate a trunked channel to a requesting user.
- **Blocked call** (lost call): a call that cannot be completed at the time of request, because of congestion.
- **Holding time**  $H$ : average duration of a typical call, in seconds.
- **Request rate**  $\lambda$ : average number of call requests per unit time.
- **Traffic intensity**  $A$ : average channel occupancy, in **Erlangs**. Dimensionless.
- **Load**: traffic intensity across the whole system.
- **1 Erlang** = one channel occupied 100 % of the time (one call-hour per hour). Eg a channel busy for 30 min in an hour carries 0.5 Erlang.

### Grade of service (GoS)

- GoS = a measure of a user's ability to access the system **during the busy hour**.
- **Busy hour** = the busiest hour of a week, month or year. For cellular it is typically 4 to 6 pm Thursday or Friday.
- Stated either as
  - **probability a call is blocked** (Erlang B), or
  - **probability a call is delayed** beyond a set queuing time (Erlang C).
- AMPS is designed for **GoS = 2 % blocking**: 2 calls in 100 are lost in the busy hour.
- The designer's job: estimate peak demand, then allocate enough channels to meet the GoS.

### The three traffic equations

- Per user:  $A_u = \lambda H$
- Total offered,  $U$  users:  $A = U A_u$
- Per channel,  $C$  channels:  $A_c = U A_u / C$
- **Offered traffic is not carried traffic**. Once offered traffic exceeds capacity, the carried traffic saturates at  $C$  Erlangs.

## The two types of trunked system

### Blocked calls cleared → Erlang B

- **No queue.** A user who finds no free channel is refused and is free to try again.
- Assumptions: Poisson arrivals, infinite users, memoryless arrivals, exponentially distributed holding time, finite channels. This is an **M/M/m/m** queue.
- GoS = probability of blocking:

$$\Pr[\text{blocking}] = \frac{\frac{A^C}{C!}}{\sum_{k=0}^C \frac{A^k}{k!}} = \text{GoS}$$

- Gives a **conservative** (pessimistic) estimate, since a finite-user model always predicts less blocking.
- In the exam you are given an **Erlang B table or chart**, never asked to sum the series.

### Blocked calls delayed → Erlang C

- **A queue holds blocked calls** until a channel frees up.
- Probability a call is delayed at all:

$$\Pr[\text{delay} > 0] = \frac{A^C}{A^C + C! \left(1 - \frac{A}{C}\right) \sum_{k=0}^{C-1} \frac{A^k}{k!}}$$

- GoS = probability the delay exceeds  $t$  seconds:

$$\Pr[\text{delay} > t] = \Pr[\text{delay} > 0] e^{-(C-A)t/H}$$

- Average delay over **all** calls:

$$D = \Pr[\text{delay} > 0] \frac{H}{C - A}$$

where  $H/(C - A)$  is the average delay of *those* calls that were queued.

**Trunking efficiency:** a large trunked group carries far more traffic per channel than several small ones at the same GoS. This is why sectoring, which splits one channel pool into 3 or 6 smaller pools, always **loses** some trunking efficiency even though it improves SIR. (81 Ba asks for exactly this loss, 6)

## 2.6 Improving coverage and capacity

### What techniques enhance capacity and coverage in a cellular radio network? 8 5 4

**PIN 1/23** 71 Ma · the sectoring Vs cell splitting comparison is **HOT 3/23** 73 Bh, 75 Bh, 81 Bh

#### Why it is needed

- A new system is deployed with few users, spread evenly → demand is low.
- As subscribers grow, demand in **some** cells exceeds what those BSs can carry.
- More spectrum is not available → capacity must come from **geometry**.

|                   | Cell splitting                                | Sectoring                                        | Microcell zone                            |
|-------------------|-----------------------------------------------|--------------------------------------------------|-------------------------------------------|
| Idea              | Subdivide a congested cell into smaller cells | Replace the omni antenna w/ directional antennas | Split a cell into zones fed by one BS     |
| $R$               | <b>Reduced</b>                                | Unchanged                                        | Unchanged                                 |
| $D/R$             | Unchanged                                     | <b>Reduced</b> (via smaller $N$ )                | Reduced                                   |
| Gains capacity by | More clusters per unit area                   | Allowing a smaller $N$                           | Allowing a smaller $N$ w/o extra handoffs |
| Handoffs          | More (smaller cells)                          | <b>Many more</b> (sector to sector)              | <b>None between zones</b>                 |
| Cost              | New BSs, towers, real estate                  | Loses trunking efficiency                        | Zone sites plus fibre or microwave links  |

## Cell splitting 5

- **Subdivide** a congested cell into smaller cells, each w/ its **own BS** and **reduced** antenna height and transmit power.
- New smaller cells are called **microcells** and are dropped in between the existing cells.
- Halving  $R$  needs **about 4 times as many cells** to cover the same area (area  $\propto R^2$ ).
- More cells  $\rightarrow$  more clusters  $\rightarrow$  more channels per unit area  $\rightarrow$  higher capacity.
- **Cell splitting merely rescales the geometry**:  $Q = D/R$  is unchanged, so the reuse plan still works.

### The power reduction (commonly examined)

- Received power must be equal at the old and new cell edges, so the reuse plan behaves the same:

$$P_r[\text{old edge}] \propto P_{t1} R^{-n}, \quad P_r[\text{new edge}] \propto P_{t2} (R/2)^{-n}$$

- Setting them equal w/  $n = 4$ :

$$P_{t2} = \frac{P_{t1}}{16} \Rightarrow \boxed{\text{reduce transmit power by 12 dB}}$$

### Problems

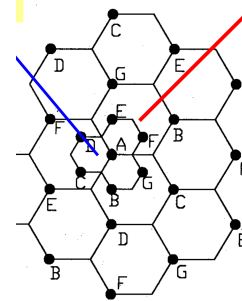
- In practice **not all cells are split at once**  $\rightarrow$  two cell sizes coexist.
- Neither the old nor the new power works for both:
  - all-high power  $\rightarrow$  small cells break the co-channel separation,
  - all-low power  $\rightarrow$  parts of the large cells go unserved.
- So channels must be **split into two groups**, one per cell size. The large cell takes high-speed traffic so it hands off less often.
- Needs new towers and antennas, and **lowers spectral efficiency** during the transition.

## Sectoring 5

- Keep  $R$  fixed, and instead attack  $D/R$ .
- Replace the single omni-directional BS antenna w/ **several directional antennas**, each radiating into one **sector**.
- A cell is normally split into **three**  $120^\circ$  or **six**  $60^\circ$  sectors.
- Each sector gets its **own** channel sub-group, used only in that sector.
- A given cell now transmits to, and receives interference from, only a **fraction** of the co-channel cells.

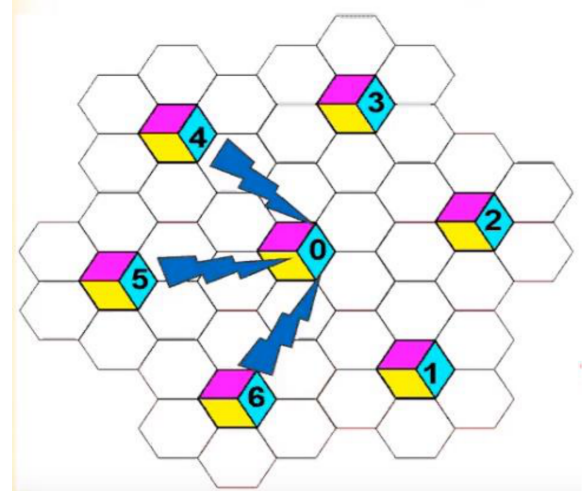
reuse plan.  
n power.

Reduce  $R$  to  $R/2$



Each original cell is surrounded by microcells of half the radius, placed so the reuse plan is preserved.

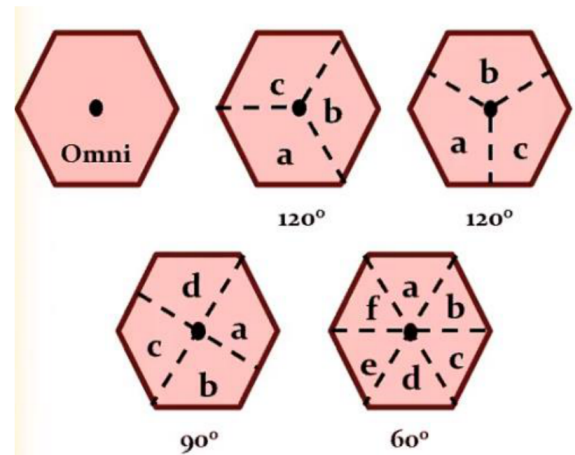
## Cell Sectoring



Omni,  $120^\circ$ ,  $90^\circ$  and  $60^\circ$  sectoring.

### The two-step gain (state it in this order)

- **Step 1:** directional antennas cut the number of first-tier interferers.
  - omni → **6** interferers,
  - $120^\circ$  → **2** interferers,
  - $60^\circ$  → **1** interferer.
- SIR therefore **improves** (17 dB → about 24.2 dB for  $120^\circ$ ).
- **Step 2:** spend that margin. Reduce  $N$  (eg  $7 \rightarrow 4$  or  $3$ ), which raises capacity, until SIR is back at the required value.
- Note the transmit power is **not** reduced. Only the relative interference is.



Sector 0 receives interference only from sectors 4, 5 and 6, so the 6 interfering cells drop to 3 (2 in the exact Rappaport treatment for  $120^\circ$ ).

### Disadvantages

- The cell's channels are split into 3 or 6 smaller pools → **trunking efficiency falls**, so carried traffic drops for the same channel count.
- **Many more handoffs**, one per sector crossing, loading the MSC.
- More antennas per site.

### Repeaters for range extension syllabus, no PYQ yet

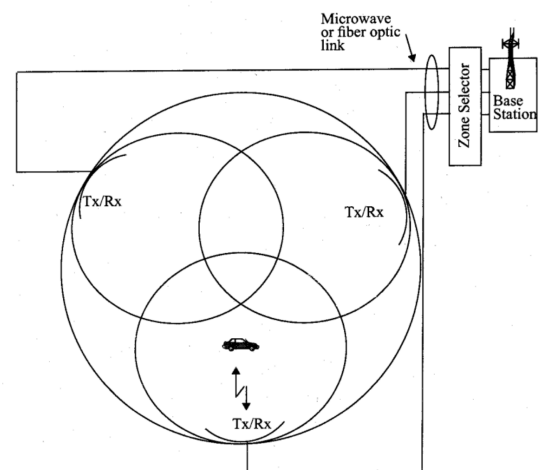
- Radio retransmitters used to cover **hard-to-reach places**: inside buildings, valleys, tunnels.
- **Bidirectional**: receives from the serving BS, amplifies, reradiates into the target area, and does the same in reverse.
- Works over the air → can be installed anywhere, repeats a whole band.
- Often feeds a **distributed antenna system (DAS)** inside a building.
- **A repeater adds coverage, not capacity**. It only reradiates the BS signal.
- It also reradiates the received **noise and interference**, so placement and gain settings need care.

### Microcell zone concept 3 1+3

HOT 3/23 80 Ba, 81 Ba, 79 Ch · asked as "**why do we need it?**", so lead with the problem

### Why it is needed (the marked part)

- Sectoring buys capacity but **multiplies handoffs**, one per sector boundary.
- Every handoff loads the MSC and the control links.
- Sectoring also **loses trunking efficiency** by splitting the channel pool.
- The microcell zone concept buys the same capacity **without either penalty**.



Three zone sites (Tx/Rx), one base station, joined by fibre or microwave. A mobile crossing between zones keeps its channel.

**How it works**

- A cell is divided into **3 or more zones**, each w/ a **zone site** (Tx/Rx) placed at the **outer edge** of the cell.
- All zone sites connect back to **one single BS**, by coaxial cable, fibre or microwave link, and share its radio equipment.
- Each zone uses a **directional antenna** radiating inward.
- The mobile is served by **whichever zone is strongest**.
- Any BS channel may be assigned to any zone.
- As the mobile crosses from zone to zone it **keeps the same channel**. **The BS just switches which zone site radiates it, so no handoff reaches the MSC.**

**Result**

- A channel is live only in the zone the mobile is in → radiation is localised → **interference falls**.
- $D/R$  drops to about **3**, so  $N$  can drop from **7 to 3**.
- Capacity rises **about 2.33 times** at the same 18 dB SIR requirement.
- Particularly good along **highways and urban traffic corridors**.

**Abbreviations used in these notes**

Only these are ever shortened. Anything not on this list is written out in full.

| Short    | Means                                   | Short    | Means                                      |
|----------|-----------------------------------------|----------|--------------------------------------------|
| BS       | base station                            | BTS      | base transceiver station                   |
| MS       | mobile station                          | MSC      | mobile switching centre                    |
| BSC      | base station controller                 | MSK      | minimum shift keying                       |
| CCI      | co-channel interference                 | ACI      | adjacent channel interference              |
| SIR, S/I | signal to interference ratio            | SNR, S/N | signal to noise ratio                      |
| GoS      | grade of service                        | MAHO     | mobile assisted handoff                    |
| FCA      | fixed channel assignment                | DCA      | dynamic channel assignment                 |
| LOS      | line of sight                           | NLOS     | non line of sight                          |
| FDMA     | frequency division multiple access      | TDMA     | time division multiple access              |
| CDMA     | code division multiple access           | OFDM     | orthogonal frequency division multiplexing |
| FDD      | frequency division duplex               | TDD      | time division duplex                       |
| GSM      | global system for mobile communications | AMPS     | advanced mobile phone system               |
| PCS      | personal communication system           | DAS      | distributed antenna system                 |
| BCCH     | broadcast control channel               | EIRP     | effective isotropic radiated power         |
| ERP      | effective radiated power                | RF       | radio frequency                            |
| Tx, TX   | transmitter                             | Rx, RX   | receiver                                   |

| Short   | Means                                       | Short | Means                                  |
|---------|---------------------------------------------|-------|----------------------------------------|
| ISI     | intersymbol interference                    | PDP   | power delay profile                    |
| DFE     | decision feedback equalizer                 | MLSE  | maximum likelihood sequence estimation |
| LMS     | least mean square                           | RLS   | recursive least squares                |
| MSE     | mean square error                           | MMSE  | minimum mean square error              |
| ZF      | zero forcing                                | MRC   | maximal ratio combining                |
| EGC     | equal gain combining                        | BER   | bit error rate                         |
| PCM     | pulse code modulation                       | DPCM  | differential PCM                       |
| ADPCM   | adaptive differential PCM                   | LPC   | linear predictive coder                |
| RPE-LTP | regular pulse excited, long term prediction | CELP  | code excited linear prediction         |
| RELTP   | residual excited linear prediction          | MPE   | multipulse excited                     |
| VSELP   | vector sum excited linear prediction        | ATC   | adaptive transform coding              |
| SBC     | sub-band coding                             | DCT   | discrete cosine transform              |
| FEC     | forward error correction                    | ARQ   | automatic repeat request               |
| BCH     | Bose–Chaudhuri–Hocquenghem code             | RS    | Reed–Solomon code                      |
| TCM     | trellis coded modulation                    | PN    | pseudo-noise                           |
| MIMO    | multiple input multiple output              | OFDMA | orthogonal FDM access                  |

| Short      | Means                                  | Short         | Means                                      |
|------------|----------------------------------------|---------------|--------------------------------------------|
| QAM        | quadrature amplitude modulation        | PSK           | phase shift keying                         |
| GMSK       | Gaussian minimum shift keying          | IMT           | international mobile telecommunications    |
| ASK        | amplitude shift keying                 | FSK           | frequency shift keying                     |
| BPSK       | binary phase shift keying              | QPSK          | quadrature phase shift keying              |
| OQPSK      | offset (staggered) QPSK                | DPSK, DBPSK   | differential (binary) PSK                  |
| CPFSK      | continuous phase FSK                   | MFSK, MPSK    | M-ary FSK, M-ary PSK                       |
| DSSS       | direct sequence spread spectrum        | FHSS          | frequency hopped spread spectrum           |
| SS, SSM    | spread spectrum (modulation)           | PG            | processing gain                            |
| PAPR       | peak-to-average power ratio            | CP            | cyclic prefix                              |
| FFT, IFFT  | (inverse) fast Fourier transform       | ICI           | inter-carrier interference                 |
| PSD        | power spectral density                 | CFO, SFO      | carrier, sampling frequency offset         |
| NRZ, RZ    | non-return, return to zero             | AMI           | alternate mark inversion                   |
| AM, FM, PM | amplitude, frequency, phase modulation | DSB, SSB, VSB | double, single, vestigial sideband         |
| S/P, P/S   | serial-parallel converter              | D/A, A/D      | digital-analog converter                   |
| LPF, BPF   | low pass, band pass filter             | ISM           | industrial, scientific, medical band       |
| SC-FDMA    | single carrier FDMA                    | <i>BT</i>     | GMSK filter bandwidth $\times$ symbol time |

| Short     | Means                                                  | Short    | Means                                               |
|-----------|--------------------------------------------------------|----------|-----------------------------------------------------|
| PSTN      | public switched telephone network                      | VoIP     | voice over IP                                       |
| FVC, RVC  | forward, reverse voice channel                         | FCC, RCC | forward, reverse control channel                    |
| SSMA      | spread spectrum multiple access                        | SDMA     | space division multiple access                      |
| FHMA      | frequency hopped multiple access                       | DSMA     | direct sequence multiple access (= CDMA)            |
| FCDMA     | hybrid FDMA/CDMA                                       | DS/FHMA  | hybrid direct sequence / frequency hopped           |
| TCDMA     | time division CDMA                                     | TDFH     | time division frequency hopping                     |
| IM        | intermodulation (product)                              | MU-MIMO  | multi-user MIMO                                     |
| WLAN      | wireless local area network                            | WMAN     | wireless metropolitan area network                  |
| WPAN      | wireless personal area network                         | AP       | access point                                        |
| BSS, IBSS | basic service set, independent (ad hoc) BSS            | ESS      | extended service set                                |
| CSMA/CA   | carrier sense multiple access with collision avoidance | CSMA/CD  | ... with collision detection                        |
| DCF       | distributed coordination function                      | PCF      | point coordination function                         |
| RTS/CTS   | request to send / clear to send                        | CCK      | complementary code keying                           |
| WEP, WPA  | wired equivalent privacy, WiFi protected access        | U-NII    | unlicensed national information infrastructure band |
| IoT       | internet of things                                     | mmWave   | millimetre wave                                     |
| eMBB      | enhanced mobile broadband                              | URLLC    | ultra-reliable low latency comm.                    |

| Short       | Means                                     | Short            | Means                                    |
|-------------|-------------------------------------------|------------------|------------------------------------------|
| mMTC        | massive machine type comm.                | NOMA             | non-orthogonal multiple access           |
| D2D         | device to device                          | C-RAN            | cloud radio access network               |
| PCN         | personal communication network            | SMS              | short message service                    |
| attr(s)     | attribute(s)                              | no.              | number                                   |
| freq        | frequency                                 | info             | information                              |
| ops         | operations                                | config           | configuration                            |
| rm          | remove                                    | vs, Vs           | versus                                   |
| w/, w/o     | with, without                             | Eg               | example / for example                    |
| +Fig        | draw a figure                             | (I)              | importance / need / significance         |
| →           | leads to / therefore                      | +                | and                                      |
| BSS         | base station subsystem                    | NSS              | network and switching subsystem          |
| OSS         | operation support subsystem               | OMC              | operation and maintenance centre         |
| HLR         | home location register                    | VLR              | visitor location register                |
| AuC         | authentication centre                     | EIR              | equipment identity register              |
| CEIR        | central EIR                               | ME               | mobile equipment                         |
| SIM         | subscriber identity module                | IMSI             | international mobile subscriber identity |
| Short       | Means                                     | Short            | Means                                    |
| IMEI        | international mobile equipment identity   | TMSI             | temporary mobile subscriber identity     |
| Um, Abis, A | the three GSM interfaces                  | SS7              | signalling system no. 7                  |
| ARFCN       | absolute radio frequency channel number   | TN, TS           | time slot number, time slot              |
| TCH         | traffic channel                           | TCH/FS, TCH/HS   | full rate, half rate speech TCH          |
| CCH         | control channel                           | CCCH             | common control channel                   |
| DCCH        | dedicated control channel                 | SDCCH            | stand-alone dedicated control channel    |
| SACCH       | slow associated control channel           | FACCH            | fast associated control channel          |
| FCCH        | frequency correction channel              | SCH              | synchronisation channel                  |
| PCH         | paging channel                            | RACH             | random access channel                    |
| AGCH        | access grant channel                      | TB, GP, SF       | tail bits, guard period, stealing flag   |
| VAD         | voice activity detector                   | DTX              | discontinuous transmission               |
| DCS 1800    | digital cellular system, the 1800 MHz GSM | CRC              | cyclic redundancy check                  |
| ESN         | electronic serial number                  | PCB              | power control bit                        |
| Mcps, Ksps  | mega-chips, kilo-symbols per second       | PRAT             | paging channel rate field                |
| EVRC        | enhanced variable rate coder              | PDSN             | packet data serving node                 |
| PCF (CDMA)  | packet control function                   | RNC              | radio network controller                 |
| Short       | Means                                     | Short            | Means                                    |
| EPS, EPC    | evolved packet system, core               | eNodeB           | LTE base station                         |
| MME         | mobility management entity                | S-GW, P-GW       | serving, packet data network gateway     |
| HSS         | home subscriber server                    | PCRF             | policy and charging rules function       |
| IMS, VoLTE  | IP multimedia subsystem, voice over LTE   | CoMP             | coordinated multipoint                   |
| ITU-R       | ITU radiocommunication sector             | WRC              | world radiocommunication conference      |
| NTA         | Nepal Telecommunications Authority        | UGS, rtPS, nrtPS | WiMAX QoS service classes                |

**Standard and system names** (proper names, kept as printed): **AMPS** advanced mobile phone system · **ETACS** extended total access communication system · **NMT** Nordic mobile telephone · **NTT** Nippon Telegraph and Telephone · **IMTS** improved mobile telephone service · **PDC** Pacific digital cellular · **IS-95** cdmaOne · **IS-136** digital AMPS · **HSCSD** high speed circuit switched data · **GPRS** general packet radio service · **EDGE** enhanced data rates for GSM evolution · **W-CDMA** wideband CDMA · **UMTS** universal mobile telecommunications system · **cdma2000 1xRTT / 1xEV-DO / 1xEV-DV** the cdma2000 3G family · **LTE, LTE-A** long term evolution (advanced) · **WiMAX** IEEE 802.16 · **WiFi** IEEE 802.11 · **DECT** digital enhanced cordless telecommunications · **CT2** cordless telephone 2 · **PHS** personal handyphone system · **USDC** US digital cellular (IS-136) · **JDC** Japanese digital cellular (later PDC) · **Bluetooth** IEEE 802.15.1 · **Zigbee** IEEE 802.15.4 · **PACS** personal access communications system · **POCSAG, FLEX** paging protocols · **CDPD** cellular digital packet

data · **Mobitex** a dedicated packet data system · **TETRA** terrestrial trunked radio · **DAB** digital audio broadcasting · **DVB-T** digital video broadcasting, terrestrial · **ADSL** asymmetric digital subscriber line · **5G NR** 5G new radio.

**Three clashes to watch:** **FCC** is the *forward control channel* throughout these notes; the US regulator is always written out as the Federal Communications Commission, and **FCCH** is GSM's frequency correction channel, a different thing again. **BCH** is the Bose–Chaudhuri–Hocquenghem block code in **Ch 6** and GSM's *broadcast channel* in **Ch 8**. **SCH** is GSM's synchronisation channel, never a sub-channel.

**Exam-paper month codes** (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

**Bold** year tags are Regular exams, plain are Back exams. Monospace year tags are BEI (EX 715); the rest are BEX (EX 751).

**Tier chips** over 23 papers, 2070–2082 BS: **TOP /23** = topic asked in 6 or more papers · **HOT /23** = 3 to 5 papers · **PIN /23** = 1 or 2 papers but worth  $\geq 6$  marks. **Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.